

Inequalities in Two Variables

AQA · KS4 Year 9–10

Name: _____ Class: _____ Date: _____

Represent linear inequalities in two variables on a graph. Draw the boundary line, decide whether it is solid or dashed, and shade the correct region. Identify the region that satisfies several inequalities together.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

Using a solid line for $<$ or $>$ — these need a DASHED line (boundary not included).

Using a dashed line for $\leq$ or $\geq$ — these need a SOLID line (boundary included).

Shading the wrong side — always test a point such as $(0, 0)$ to decide.

Choosing a test point that lies on the line, which cannot tell you which side to shade.

METHOD

- 1 Replace the inequality sign with $=$ and draw that straight line.
- 2 Use a dashed line for $<$ or $>$, and a solid line for $\leq$ or $\geq$.
- 3 Choose a test point not on the line (usually $(0, 0)$) and substitute it in.
- 4 If the point makes the inequality true, shade its side; if not, shade the other side.
- 5 For several inequalities, the required region is where all the shaded areas overlap.

WORKED EXAMPLE

Show the region for: $y \leq x + 2$

Step 1 — Draw the line $y = x + 2$.

Step 2 — $\leq$ means the line is included, so draw it SOLID.

Step 3 — Test $(0, 0)$: is $0 \leq 0 + 2$? Yes, $0 \leq 2$.

Step 4 — $(0, 0)$ works, so shade the side that contains $(0, 0)$ (below the line).

Step 5 — (Single inequality, so the shaded region is the answer.)

1 State whether the boundary line should be solid or dashed.

(2 marks)

a) $y > 2x$

b) $y \leq x + 1$

HOW TO START

$<$ and $>$ use a dashed line; $\leq$ and $\geq$ use a solid line.

Look only at the inequality sign to decide.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Draw the boundary line (sign $=$) | 2 Dashed for $</>$, solid for $\leq/\geq$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

2 Decide whether each point satisfies the inequality.

(2 marks)

a) Is (1, 4) in the region $y > x + 2$?

b) Is (2, 1) in the region $y \geq 2x - 1$?

HOW TO START

Substitute the point's x and y values into the inequality.

If the statement is true, the point is in the region.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Draw the boundary line (sign $\neq$) | 2 Dashed for $</>$, solid for $\leq / \geq$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

(3 marks)

c) $y < x$

METHOD 1 Draw the boundary line (sign =) | 2 Dashed for $</>$, solid for $\leq / \geq$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

4 By testing the point (0, 0), state whether (0, 0) lies in each region.

(3 marks)

a) $y < x + 1$

b) $y \geq 2x - 3$

c) $y > x$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Draw the boundary line (sign $\neq$) | 2 Dashed for $</>$, solid for $\leq / \geq$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

5 On one set of axes, shade the region R that satisfies all three inequalities.

(4 marks)

$$x \geq 0, y \geq 0, x + y \leq 4$$

State the coordinates of the three corners of the region.

ANSWER

Answer =

METHOD 1 Draw the boundary line (sign =) | 2 Dashed for $</>$, solid for $\leq / \geq$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

- 6** A region is bounded by the lines $x = 1$, $y = 1$ and $y = 5 - x$, and lies above $y = 1$, to the right of $x = 1$ and below $y = 5 - x$. (3 marks)

Write the three inequalities that describe this region.

ANSWER

Answer = _____

METHOD 1 Draw the boundary line (sign $=$) | 2 Dashed for $</>$, solid for $/$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

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(4 marks)

a) State which boundary lines should be solid. (1 mark)

b) The point $(3, 2)$ — is it inside region R? Show your working. (3 marks)

ANSWERS

(a) = _____

(b) =

METHOD 1 Draw the boundary line (sign =) | 2 Dashed for $</>$, solid for $/$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

8 A delivery van carries x small parcels and y large parcels. (5 marks)
The rules are: $x + y \leq 10$, $x \geq 2$, $y \geq 3$.

a) Write what each inequality means in words. (2 marks)

b) Give one possible combination (x, y) that satisfies all three rules. (1 mark)

c) What is the greatest total number of parcels the van can carry, and give an (x, y) that achieves it. (2 marks)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Draw the boundary line (sign $=$) | 2 Dashed for $</>$, solid for $\leq / \geq$ | 3 Test a point not on the line | 4 Shade the side that works | 5 Several inequalities shade the overlap

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet

☐ Getting there

☐ Confident

☐ Really confident

Which question did you find the hardest, and why?

How do I decide which side of the line to shade?

One thing I will practise before the next lesson:

Teacher key

Q1: a) dashed ($>$) b) solid ()

Q2: a) $4 > 1 + 2 = 3$, true yes b) $1 - 2(2) - 1 = 3$, true yes

Q3: a) dashed vertical line $x = 2$, shade right b) solid horizontal line $y = 3$, shade below c) dashed line $y = x$, shade below it

Q4: a) $0 < 0 + 1 = 1$, true yes b) $0 - 2(0) - 3 = -3$, true yes c) $0 > 0$ is false no

Q5: triangle with corners $(0, 0)$, $(4, 0)$ and $(0, 4)$; lines $x = 0$, $y = 0$, $x + y = 4$ all solid; shade inside

Q6: $x \geq 1$, $y \geq 1$, $y \leq 5 - x$

Q7: a) all three are solid (, ,) b) $y \geq 1$; $2x - 1 \geq 2$; $2(3) - 1 = 5 \geq 2$; $x \leq 4$; $3 \leq 4$ — all true, so $(3, 2)$ is inside R

Q8: a) total parcels at most 10; at least 2 small; at least 3 large b) e.g. $(3, 4)$ since $3 + 4 = 7 \leq 10$, $3 \geq 2$, $4 \geq 3$ c) greatest total = 10, e.g. $(2, 8)$ or $(7, 3)$ [$x + y = 10$ with $x \geq 2$, $y \geq 3$]
